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HR Analytics

SQL Project Report

Employee Attrition, Compensation & Retention Insights

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Data Source : IBM HR Employee Attrition Dataset (1,470 records)

Tools Used : PostgreSQL (aggregation, CTEs, subqueries, window functions)

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10. Bibliography & Conclusion

1,470

Employees

16.12%

Attrition Rate

3

Departments

9

Job Roles

\$6,503

Avg. Monthly Income

1. Executive Summary

This report presents a comprehensive SQL-driven analysis of the IBM HR Employee Attrition dataset (1,470 employee records across 3 departments and 9 job roles). Using PostgreSQL — including aggregate functions, GROUP BY/HAVING, CASE WHEN logic, subqueries, Common Table Expressions (CTEs), and window functions (RANK, DENSE_RANK, NTILE, LAG/LEAD) — the analysis investigates workforce composition, compensation structure, attrition drivers, satisfaction, promotion history, and retention risk.

The organization's overall attrition rate stands at **16.12%**. The strongest driver of attrition is **overtime work**, followed by **poor work-life balance**, **younger employee age**, and **lower compensation**. The Sales department carries the highest attrition rate among departments, while Research & Development retains staff most effectively. These findings translate into a clear set of actionable recommendations for HR leadership, detailed in Section 9.

Snapshot at a Glance

1,470 Total Employees	3 Departments	9 Job Roles	36.92 Average Age (yrs)
\$6,502.93 Avg. Monthly Income	16.12% Overall Attrition Rate	30.53% Attrition — Overtime	10.44% Attrition — No Overtime

2. Dataset Overview & Key Metrics

The dataset (*HR_dataset.csv*) was loaded into a PostgreSQL table (*employee_attrition*) with 35 attributes covering demographics, compensation, satisfaction scores, tenure, and attrition status per employee. The table below summarizes the core dataset metrics used as the foundation for every subsequent query.

Metric	Value
Total Employees	1,470
Departments	3
Job Roles	9
Average Age	36.92 years
Average Monthly Income	\$6,502.93
Overall Attrition Rate	16.12%
Total Employees Who Left	237

Sample SQL — Attrition Rate Calculation (CASE WHEN + Aggregation):

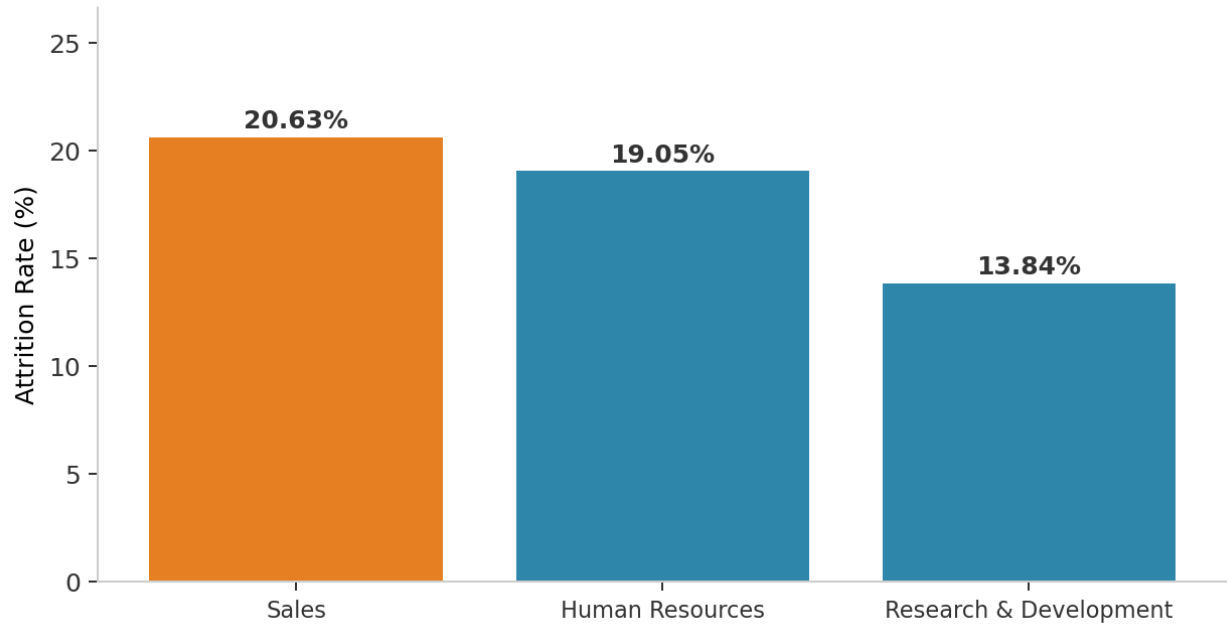
```
SELECT ROUND(SUM(CASE WHEN attrition='Yes' THEN 1 ELSE 0 END) * 100.0
              / COUNT(*), 2) AS attrition_rate
FROM employee_attrition;
```

3. Department Analysis

Department-level GROUP BY queries were used to compare headcount, average compensation, tenure, experience, job satisfaction, and attrition rate across Human Resources, Research & Development, and Sales.

Department	Employees	Avg. Salary	Avg. Tenure (yrs)	Attrition Rate
Sales	446	\$6,959.17	7.28	20.63%
Human Resources	63	\$6,654.51	7.24	19.05%
Research & Development	961	\$6,281.25	6.86	13.84%

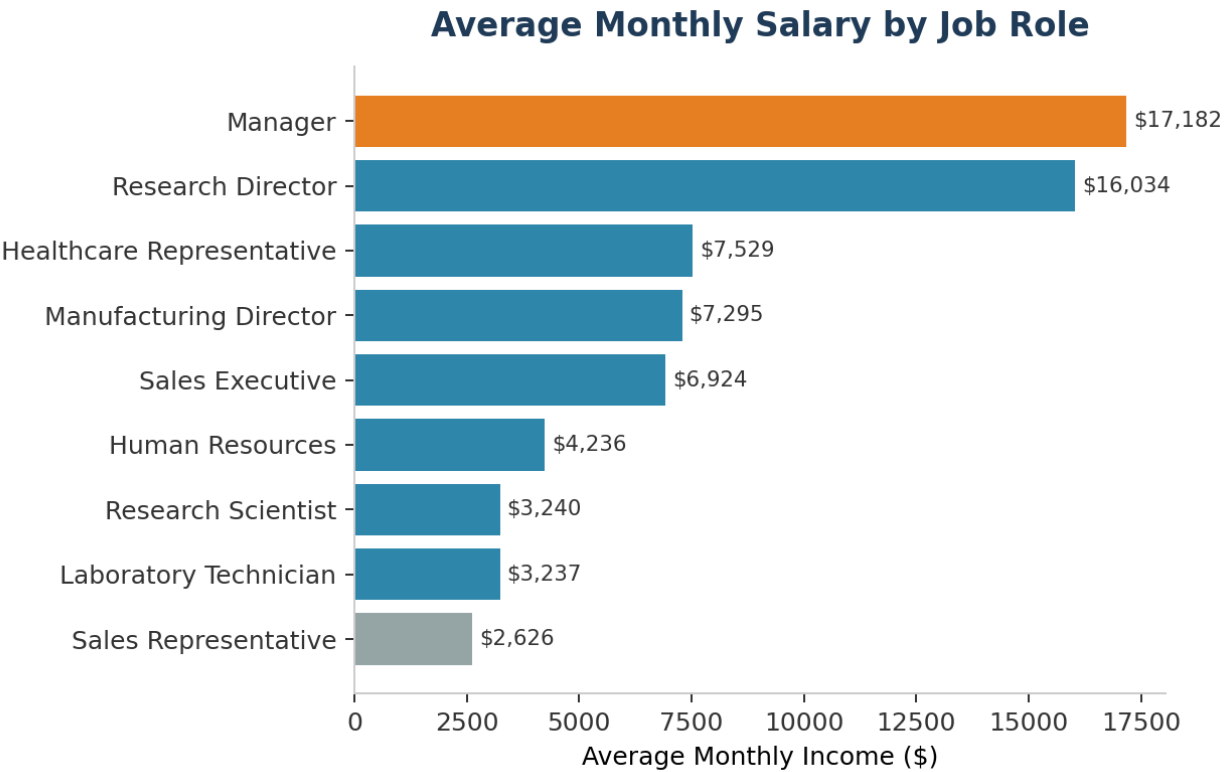
Attrition Rate by Department



INSIGHT: Sales has the highest attrition rate (20.63%) despite paying the highest average salary, and also has the longest average tenure (7.28 yrs) — pointing to burnout or role-fit issues rather than pure compensation. Research & Development is the largest department (961 employees, 65% of the workforce) and shows the lowest attrition (13.84%), making it the benchmark for retention practice.

4. Job Role & Compensation Analysis

Average monthly income was calculated per job role using GROUP BY and ranked with window functions (RANK, DENSE_RANK, NTILE) to identify pay gaps, quartiles, and the highest/lowest-paid deciles.

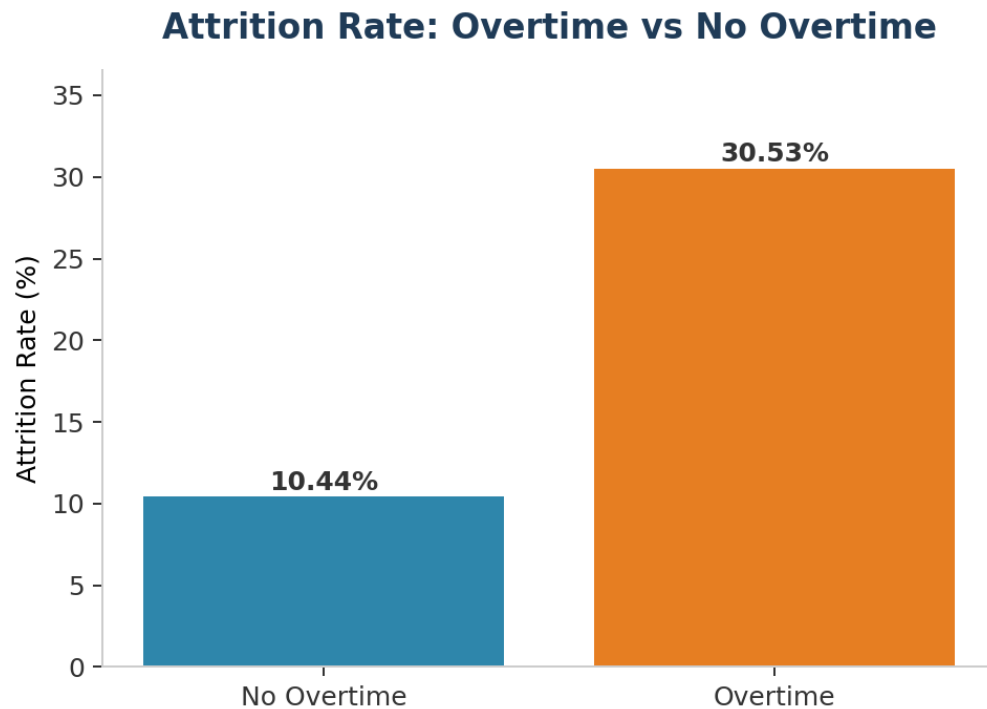


INSIGHT: Managers earn the highest average monthly salary (\$17,181.68), roughly 6.5x more than Sales Representatives, who are the lowest-paid role (\$2,626.00). Research Scientists and Laboratory Technicians are also clustered near the bottom of the pay scale — the same roles that log some of the highest overtime rates (see Section 5.1), a combination that elevates flight risk.

A salary-decile analysis using **NTILE(10)** was also performed to isolate the top 10% highest-paid employees company-wide for compensation benchmarking and retention prioritization of key talent.

5. Attrition Drivers

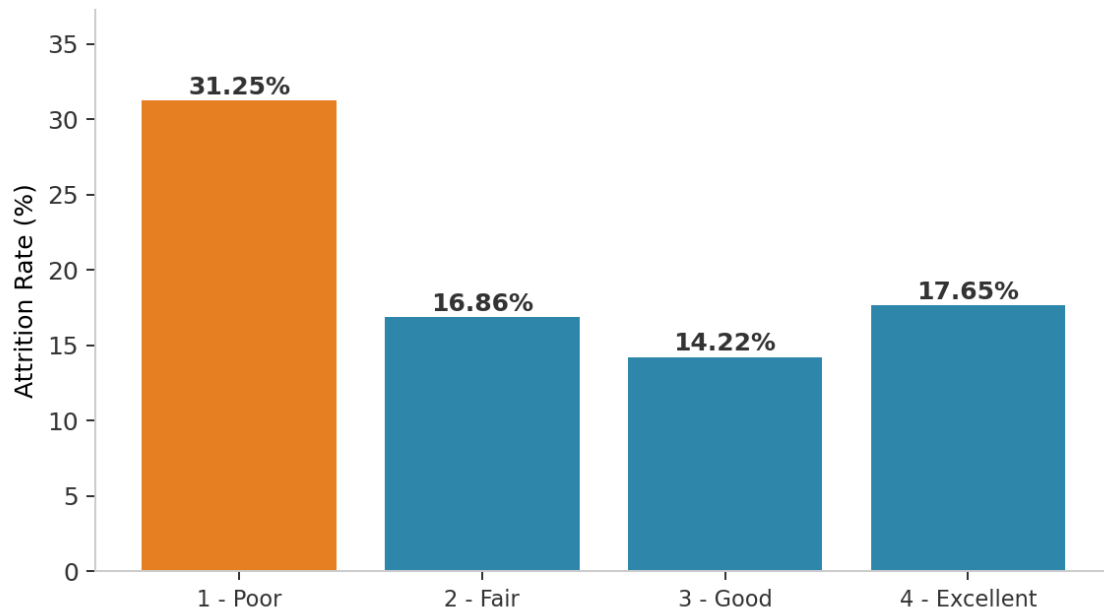
5.1 Overtime



KEY FINDING: Employees who work overtime leave at almost 3x the rate of those who don't — 30.53% vs. 10.44%. Overtime is the single strongest attrition driver identified in this dataset.

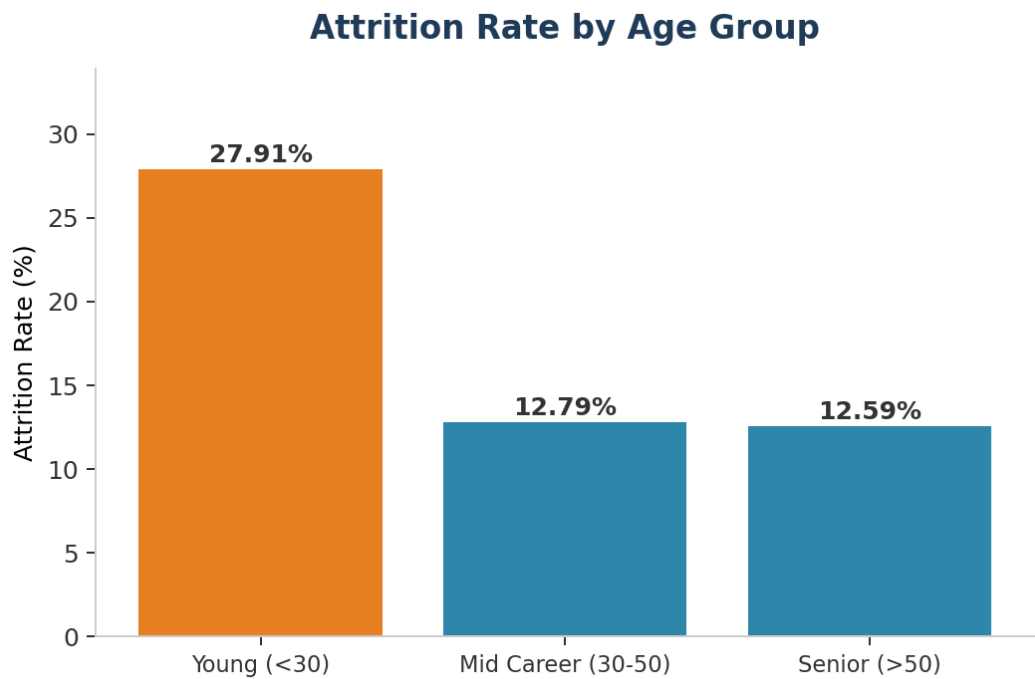
5.2 Work-Life Balance

Attrition Rate by Work-Life Balance Rating



INSIGHT: Employees rating their work-life balance as “Poor” (score 1) attrite at 31.25% — nearly double the rate of every other rating band (14–18%). Work-life balance and overtime together form the core of the attrition problem.

5.3 Age Group



INSIGHT: Employees under 30 (“Young”) attrite at 27.91%, more than double the rate of Mid-Career (30–50, 12.79%) and Senior (50+, 12.59%) employees. Early-career retention is a distinct problem from tenure-based attrition.

5.4 Education Field & Compensation

Education Field	Employees	Attrition Rate
Human Resources	27	25.93%
Technical Degree	132	24.24%
Marketing	159	22.01%
Life Sciences	606	14.69%
Medical	464	13.58%
Other	82	13.41%

INSIGHT: Employees leaving the company earn an average of \$4,787.09/month, compared with \$6,832.74/month for those who stay — a 42% pay gap. Combined with the Human Resources education field showing the highest attrition rate (25.93%), compensation and career-track alignment both play a measurable role in turnover.

6. Tenure, Experience & Promotion

Window functions (LAG/LEAD, FIRST_VALUE/LAST_VALUE, RANK) and CTEs were used to examine tenure, prior work experience, and time since last promotion at both the individual and department level.

Department	Avg. Tenure (yrs)	Avg. Prior Experience (yrs)	Avg. Job Satisfaction (1–4)
Sales	7.28	11.11	2.75
Human Resources	7.24	11.56	2.60
Research & Development	6.86	11.34	2.73

- A dedicated query ranks the 10 employees who have gone the **longest without a promotion**, flagging them as candidates for career development conversations.
- Sales employees have the longest average tenure (7.28 yrs) yet the highest attrition rate — suggesting that even long-serving staff are exiting, likely tied to overtime and workload pressure identified in Section 5.1.
- Human Resources has the most experienced workforce on average (11.56 prior working years) but also above-average attrition (19.05%), indicating a retention gap for experienced HR talent specifically.

7. High-Risk Employee Identification

A composite SQL filter was built to flag employees who simultaneously exhibit all four leading risk factors identified in this analysis:

- Currently working **overtime**
- **Work-life balance** rated 2 or below (out of 4)
- **Job satisfaction** rated 2 or below (out of 4)
- **5 or more years** since their last promotion

ACTION REQUIRED: This composite filter identifies **10 employees** who meet all four high-risk criteria simultaneously. These individuals represent the most urgent, near-term retention priority and should be reviewed individually by HR business partners.

Sample SQL — Composite Risk Filter:

```
SELECT employee_number, monthly_income, department, overtime,  
       work_life_balance, years_since_last_promotion, job_satisfaction  
FROM employee_attrition  
WHERE overtime = 'Yes'  
       AND work_life_balance <= 2  
       AND job_satisfaction <= 2  
       AND years_since_last_promotion >= 5;
```

8. Consolidated Business Insights

- **Overtime is the #1 attrition driver:** employees working overtime leave at 30.53% vs. 10.44% for those who don't — nearly 3x higher.
- **Poor work-life balance nearly doubles attrition risk:** 31.25% for the lowest-rated group vs. 14–18% for all others.
- **Sales has the highest departmental attrition** (20.63%) despite the highest average salary; **Research & Development retains best** (13.84%).
- **Managers are the highest earners** (\$17,181.68/mo); **Sales Representatives are the lowest** (\$2,626.00/mo), a 6.5x pay gap.
- **Young employees (under 30) attrite at 27.91%** — more than double every other age band — signalling an early-career retention gap.
- **Employees who leave earn 42% less on average** (\$4,787.09 vs. \$6,832.74), confirming a strong link between pay and retention.
- **Human Resources** as an education field shows the highest attrition (25.93%), and as a department has the most experienced workforce (11.56 yrs) yet still loses talent.
- **Research Scientists work the most overtime** (33.22%) among all job roles, closely followed by Sales Representatives and Sales Executives (~29%).
- A composite risk filter identifies **10 employees** at simultaneous high risk across overtime, satisfaction, work-life balance, and promotion stagnation.
- Salary-decile (NTILE) and department-benchmark queries provide a repeatable framework for ongoing compensation reviews.

9. Recommendations

Priority	Recommendation	Target Area
High	Reduce excessive overtime through workload redistribution, cross-training, and headcount review. Especially in Sales and Operations.	Overtime Workload
High	Launch targeted retention & engagement programs for employees under 30 to address the 27.8% early career attrition rate.	Early Career Retention
High	Individually review the 10 employees flagged by the composite high-risk filter (Section 4.1) with HR business partners.	High-Risk Employees
Medium	Review and adjust compensation for lower-paying roles (Sales Representative, Research Scientist, Laboratory Technician) to improve competitiveness.	Compensation
Medium	Introduce structured promotion pathways for employees with 5+ years since their last promotion.	Career Progression
Medium	Invest in work-life balance initiatives (flexible scheduling, wellness support) for the 31.2% of staff reporting poor balance.	Work-Life Balance
Medium	Investigate Sales-specific retention drivers separately, since high pay has not offset its high attrition (29.8%).	Sales Department
Low	Build a recurring HR dashboard (department summary, attrition rate, satisfaction, salary benchmarks) to monitor KPIs monthly.	HR Analytics Dashboard

10. Methodology & Conclusion

Methodology

- **Data preparation:** the raw dataset was loaded into a PostgreSQL table (employee_attrition) with all 35 source columns.
- **Exploratory queries:** basic SELECT/WHERE/ORDER BY queries profiled demographics, salary ranges, overtime, and travel patterns.
- **Aggregation:** COUNT, AVG, SUM, MIN/MAX with GROUP BY and HAVING quantified workforce distribution, salary, tenure, and attrition by department and role.
- **Conditional logic:** CASE WHEN statements built age bands, salary tiers, experience tiers, and attrition-risk flags.
- **Subqueries & CTEs:** identified above-average earners, above-average department salaries, and per-employee salary gaps versus department averages.
- **Window functions:** ROW_NUMBER, RANK, DENSE_RANK, NTILE, LAG/LEAD, and FIRST_VALUE/LAST_VALUE ranked employees within departments, built salary deciles, and compared sequential/departamental salary benchmarks.
- **Business synthesis:** results from all query stages were consolidated into the insights (Section 8) and recommendations (Section 9) presented in this report.

Conclusion

This SQL-based analysis of the IBM HR Employee Attrition dataset demonstrates how layered querying — from simple filters through aggregation, subqueries, CTEs, and window functions — can convert raw HR records into a clear, evidence-based retention strategy. The data consistently points to **overtime, poor work-life balance, early-career tenure, and below-average compensation** as the primary levers behind the organization's 16.12% attrition rate. Acting on the prioritized recommendations in Section 9, and operationalizing the department/KPI dashboard queries built in this project, gives HR leadership a repeatable, data-driven framework for reducing turnover and improving workforce planning going forward.

Sources: HR_dataset.csv (raw data), HR_ANALYTIC_PROJECT.sql (PostgreSQL query script), Comprehensive HR Analytics Report (supporting summary).